

Improving Precision & Power in Randomized Trials Using Covariate Adjustment

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Recap & Notation

High-Level Recap on Covariate Adjustment

- Prognostic factors X ; Measured before treatment assignment A ;
- **Variance of Chance Imbalances** → Variance of unadjusted estimates
 - Depends on Variance in Population; Inclusion/Exclusion; Design
 - $I = (\sum_{i=1}^N A_i X_i / \sum_{i=1}^N A_i) - (\sum_{i=1}^N (1 - A_i) X_i / \sum_{i=1}^N (1 - A_i))$
 - $\uparrow \text{var}(I) \rightarrow \uparrow \text{var}(\hat{\theta}_U)$ - Unadjusted estimate
- Design: stratified randomization - limits potential imbalance
- Analysis: Covariate Adjustment - Complement to design
 - Conditional (*model-based*): Correctly specified - *a priori*
 - Marginal (*model-assisted*): Working model not assumed to be correctly specified

Marginal Approach:

- 1 Estimate same quantity (estimand) as unadjusted analysis: θ
 - 2 Same or weaker assumptions
 - 3 Covariates are *prognostic*: $\uparrow$ Precision $\uparrow$ Power
- Adjustment improves precision by addressing chance imbalances in X using *working models* that aren't assumed to be correctly specified. Models used for prediction, not interpretation.
 - Randomization $\rightarrow A \perp X \rightarrow$ robustness to misspecification.
 - Not guaranteed: Observational; Causal Inference methods
 - Benefit: $var(I)$ - Gain *even if allocation is balanced*
 - Difference: $\hat{\theta}_A - \hat{\theta}_U$ depends on observed imbalance

Terminology, Notation, and Assumptions

- Observed outcome Y , treatment assignment A , baseline covariates X
 - $A = 1$: Treatment; $A = 0$: Control
- N Participants: $(X_i, A_i, Y_i) \stackrel{\text{iid}}{\sim} P$ where P is unknown
- Treatment assigned independently of X : $A \perp X$ in RCT
 - Stratified Randomization: $X = (X_S, X_{\bar{S}})$ then $A \perp\!\!\!\perp X_{\bar{S}} | X_S$
- Same/similar assumptions needed for consistency of unadjusted estimator
- Consistency of estimator: $\hat{\theta} \xrightarrow{p} \theta$ i.e. converges asymptotically to true value of the estimand

Hands-On

Learning Objectives: Hands-On

- Identify packages for adjusted analyses
- Identify resources for self-learning: tutorials, data
- Compute adjusted analyses for continuous, binary, ordinal, time-to-event outcomes
- Be able to estimate precision gain: historical data

Hands-On

- Focus on R: (+) Free & Open Source; (+) Capabilities; (-) Syntax
 - Stata: `teffects`; SAS Macros: DIY
 - New to R? See [Learning & Using R section](#).
- What > How: Software generally fairly straightforward
 - Regression: Outcome Y , Treatment assignment A , Covariates X
 - Estimand, Design (Scheme, Strata), Variance Estimation
 - Lists of variable names (SAS, Stata, R) vs. Formulas (R)

Using R

■ Download R and R Studio

- Windows: [RTools](#) - Compilers; Built into iOS/Linux; **Only when needed.**

- Building packages from source code vs. pre-compiled binary files: Source may have newer version than binaries.

■ Download the .html - Easier to follow the code

- Code can be copied by clicking upper right corner

■ Download the .qmd - Run chunk-by-chunk

■ Install packages: `install.packages` (CRAN) `pak::pak()` (GitHub)

Software:

■ Several R packages on CRAN and GitHub (GH): Outcomes, Estimands

Package	Continuous	Binary	Ordinal	Time-to-Event	Repo
adjrct			✓	✓	GH
AIPW	✓	✓		✓	CRAN
drord			✓		GH
PSW	✓	✓			CRAN
RobinCar	✓	✓		✓	CRAN
RobinCar2	✓	✓		✓	CRAN
speff2trial	✓	✓		✓	CRAN
survtmle				✓	GH

■ Need for software that meets standards of regulatory guidance

■ Validation plan; Testing workflow; Code coverage; Repository state

Example RCT Datasets

- [examplercts package](#): 9 analysis-ready datasets: Dictionaries, Variable/Value Labels

Dataset	Continuous	Binary	Ordinal	Time-to-Event	Longitudinal
strep_tb		✓	✓		
indo_rct		✓			
botox_dystonia	✓				✓
laryngoscope	✓	✓	✓	✓	
licorice_gargle	✓	✓	✓		✓
supraclavicular	✓	✓	✓	✓	
perio_pregnancy	✓	✓	✓		✓
colon_cancer				✓	
lung_cancer				✓	

- [RCT Bench](#) - 125 completed trials: Searchable, Dictionaries ([Shao et al. 2026](#))

CovariateAdjustment.GitHub.io

- Example Data & Code:
 - Tabulations, figures, working with R
- Estimands and estimation methods
- Survey of literature; Practical guidance
- Major update: Next two months
 - Missing Data; Double Robustness; Ordinal, Longitudinal analyses
- Future: Cluster RCTs, Information Monitoring, New methods

Installing Packages

```
install.packages(  
  pkgs = c("dplyr", "drord", "cobalt", "kableExtra", "lmtest",  
           "pak", "sandwich", "speff2trial", "table1")  
)  
  
# Installing packages from GitHub  
pak::pak(  
  pkg =  
    c("covariateadjustment/examplercts",  
      "jbetz-jhu/drwls",  
      "nt-williams/simul",  
      "nt-williams/adjrct")  
)
```

Load Packages

```
# Continuous, Binary Analyses
library(dplyr) # Wrangling data
library(drwls) # G-computation & DR-WLS
library(cobalt) # Standardized Differences
library(examplercts) # Example Datasets
library(kableExtra) # Printing Tables in HTML/TeX
library(lmtest) # Computing CIs with Robust SEs
library(sandwich) # Computing Robust SEs for GLMs
library(table1) # Tabulations

# Ordinal, Time-to-Event Analyses
library(adjrct) # Time-to-Event: Survival & RMST
library(drord) # Adjusted Ordinal Analyses
library(speff2trial) # Adjusted Marginal Hazard Ratio
library(survival) # Logrank tests; Cox PH Model
```

licorice_gargle Data in exemplercats Package

- Licorice gargle (Treatment) vs. Sucrose Gargle (Control)
- Outcomes: Cough, Throat Pain at 30m, 90m, 4h post-extubation
 - Pain: 0 (None) - 10 (Worst Imaginable)
 - Treat as continuous and binary (0 vs. 1-10); Ordinal
 - Cough: No Cough < Mild < Moderate < Severe
- Covariates: Age, Sex, BMI, Physical Status, Mallampati Score

Tabulations & Balance:

- `table1::table1()` - Producing HTML/LaTeX summary tables

```
table1::table1(  
  x = ~ age_bl + gender + asa_physical_status_bl +  
    bmi_bl + mallampati_bl + pain_yn_bl | arm,  
  data = licorice_gargle  
)
```

- `cobalt::bal.tab` - Calculating Standardized Differences

```
cobalt::bal.tab(  
  x =  
    licorice_gargle %>%  
    dplyr::select(dplyr::ends_with(match = "_bl")),  
  treat = licorice_gargle$arm,  
  binary = "std",  
  continuous = "std",  
  s.d.denom = "pooled"  
)
```


Baseline Covariates

■ Missingness; Low Frequency Categories; Balance

	0. 5g Sugar (N=117)	1. 0.5g Licorice (N=118)	Overall (N=235)
Age at Baseline			
Mean (SD)	58.0 (16.1)	56.7 (14.9)	57.4 (15.5)
Median [Min, Max]	63.0 [18.0, 86.0]	60.5 [19.0, 83.0]	62.0 [18.0, 86.0]
Gender			
0. Female	44 (37.6%)	49 (41.5%)	93 (39.6%)
1. Male	73 (62.4%)	69 (58.5%)	142 (60.4%)
ASA Physical Status			
1. Healthy	19 (16.2%)	22 (18.6%)	41 (17.4%)
2. Mild Systemic Disease	67 (57.3%)	67 (56.8%)	134 (57.0%)
3. Severe Systemic Disease	31 (26.5%)	29 (24.6%)	60 (25.5%)
Body Mass Index (kg/m²)			
Mean (SD)	25.6 (4.25)	25.6 (4.32)	25.6 (4.27)
Median [Min, Max]	26.1 [15.6, 34.1]	25.7 [16.4, 36.3]	25.9 [15.6, 36.3]
Mallampati Score (BL)			
1. 1	31 (26.5%)	39 (33.1%)	70 (29.8%)
2. 2	69 (59.0%)	66 (55.9%)	135 (57.4%)
3. 3+	17 (14.5%)	13 (11.0%)	30 (12.8%)
Preoperative Pain (BL)			
0. No	115 (98.3%)	118 (100%)	233 (99.1%)
1. Yes	2 (1.7%)	0 (0%)	2 (0.9%)

Standardized Differences: Imbalance

```
## Balance Measures
##
## age_bl                               Type Diff.Un
## asa_physical_status_bl_1. Healthy    Binary  0.0634
## asa_physical_status_bl_2. Mild Systemic Disease Binary -0.0098
## asa_physical_status_bl_3. Severe Systemic Disease Binary -0.0440
## bmi_bl                               Contin. -0.0122
## mallampati_bl_1. 1                   Binary  0.1437
## mallampati_bl_2. 2                   Binary -0.0616
## mallampati_bl_3. 3+                  Binary -0.1054
## smoking_bl_1. Current Smoker         Binary -0.0067
## smoking_bl_2. Former Smoker          Binary -0.0057
## smoking_bl_3. Never Smoker           Binary  0.0127
## pain_yn_bl_1. Yes                    Binary -0.1865
##
## Sample sizes
##      0. 5g Sugar 1. 0.5g Licorice
## All      117      118
```

■ Standardized Difference > 0.10: Correlation with outcome

Unadjusted Analysis: t-Test

```
licorice_30_min_pain_t_test <-  
  t.test(  
    pacu_30_min_throat_pain ~ arm, data = licorice_gargle  
  )  
licorice_30_min_pain_t_test  
##  
## Welch Two Sample t-test  
##  
## data: pacu_30_min_throat_pain by arm  
## t = 4.8035, df = 157.3, p-value = 3.608e-06  
## alternative hypothesis: true difference in means between group 0. 5g Sugar and g  
## 95 percent confidence interval:  
## 0.4429931 1.0617225  
## sample estimates:  
## mean in group 0. 5g Sugar mean in group 1. 0.5g Licorice  
## 1.0258621 0.2735043  
# Note: t-test is presenting [Control] (Reference) - [Treatment]  
diff(licorice_30_min_pain_t_test$estimate)  
## mean in group 1. 0.5g Licorice  
## -0.7523578  
-rev(licorice_30_min_pain_t_test$conf.int)  
## [1] -1.0617225 -0.4429931
```

ANCOVA - Code

```
licorice_30_min_pain_glm_adjusted <-  
  glm(  
    pacu_30_min_throat_pain ~ arm +  
      age_bl + gender + asa_physical_status_bl + bmi_bl + mallampati_bl,  
    data = licorice_gargle,  
    family = gaussian  
  )  
  
# Robust Huber-White SEs  
licorice_30_min_pain_ancova_robust_vcov <-  
  sandwich::vcovHC(licorice_30_min_pain_glm_adjusted, type = "HCO")  
  
# CIs based on Robust SEs - lmtest::coeftest for Tests  
licorice_30_min_pain_ancova_robust_ci <-  
  lmtest::coefci(  
    x = licorice_30_min_pain_glm_adjusted,  
    vcov. = licorice_30_min_pain_ancova_robust_vcov  
  )
```

ANCOVA - Results

```
##
## z test of coefficients:
##
##
```

	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	1.3557577	0.5555588	2.4403	0.01467 *
## arm1. 0.5g Licorice	-0.7219391	0.1510823	-4.7784	1.767e-06 ***
## age_bl	-0.0044495	0.0062754	-0.7090	0.47830
## gender1. Male	0.2889391	0.1538593	1.8779	0.06039 .
## asa_physical_status_bl2. Mild Systemic Disease	0.2428688	0.2128941	1.1408	0.25395
## asa_physical_status_bl3. Severe Systemic Disease	0.3339201	0.2635809	1.2669	0.20521
## bmi_bl	-0.0264722	0.0192630	-1.3743	0.16936
## mallampati_bl2. 2	0.2889037	0.1733967	1.6661	0.09568 .
## mallampati_bl3. 3+	0.1828057	0.2619940	0.6977	0.48534

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

	2.5 %	97.5 %
## (Intercept)	0.26688241	2.444633053
## arm1. 0.5g Licorice	-1.01805508	-0.425823178
## age_bl	-0.01674907	0.007850052
## gender1. Male	-0.01261948	0.590497703
## asa_physical_status_bl2. Mild Systemic Disease	-0.17439596	0.660133622
## asa_physical_status_bl3. Severe Systemic Disease	-0.18268894	0.850529212
## bmi_bl	-0.06422693	0.011282550
## mallampati_bl2. 2	-0.05094757	0.628754917
## mallampati_bl3. 3+	-0.33069311	0.696304498

G-computation from GLM

```
# Step 2: Use working model to predict outcomes under each treatment
pred_30_min_pain_licorice <-
  predict(
    object = licorice_30_min_pain_glm_adjusted,
    newdata =
      within(
        data = licorice_gargle,
        expr = {arm = "1. 0.5g Licorice"}
      ),
    type = "response"
  )

pred_30_min_pain_sucrose <-
  predict(
    object = licorice_30_min_pain_glm_adjusted,
    newdata =
      within(
        data = licorice_gargle,
        expr = {arm = "0. 5g Sugar"}
      ),
    type = "response"
  )

# Step 3: Average predictions
mean_pred_30_min_pain_licorice <- mean(pred_30_min_pain_licorice)
mean_pred_30_min_pain_sucrose <- mean(pred_30_min_pain_sucrose)

# Step 4: Contrast Predictions
difference_licorice_sucrose <-
  mean_pred_30_min_pain_licorice - mean_pred_30_min_pain_sucrose
```

G-computation: `drwls::standardization`

```
licorice_30_min_pain_g_computation <-  
  drwls::standardization(  
    outcome_formula =  
      pacu_30_min_throat_pain ~ tx + age_bl + gender +  
      asa_physical_status_bl + bmi_bl + mallampati_bl,  
    outcome_family = gaussian,  
    treatment_column = "tx",  
    estimand = "difference", # Compute Difference in Means  
    data = licorice_gargle,  
    se_method = "asymptotic" # "bootstrap"  
  )
```

G-computation: drwls::standardization

```
##
## Estimand: Difference in Means
## Method: Marginal Standardization (G-Computation)
## Outcome Models: a gaussian GLM with an identity link
##
##
##           Estimand   Estimate      SE      LCL      UCL      Z      p-value
## 1 E[Y|A=1] - E[Y|A=0] -0.7219391 0.1510823 -1.018055 -0.4258232 -4.778448 1.766534e-06
## 2           E[Y|A=1]  0.2887267      NA      NA      NA      NA      NA
## 3           E[Y|A=0]  1.0106658      NA      NA      NA      NA      NA
##
## Variance Estimation: Sandwich Estimator
## CI Method: Wald 95% CI
## Details:
## Outcomes: (N = 233/235)
## Outcome Model: pacu_30_min_throat_pain ~ tx + age_bl + gender + asa_physical_status_bl + bmi_bl + ma
```

- Detailed but compact output: Estimand; Method; Working Model; Methods for SE + CI; Missing Data; Formulas
- Easy to extract high-level components (estimates, SE, CI, p -values) and low level components (models, predictions, variance adjustments)

ANCOVA vs. G-computation

```
diff(licorice_30_min_pain_t_test$estimate) # T-test
## mean in group 1. 0.5g Licorice
## -0.7523578
difference_licorice_sucrose # G-computation: Manual
## [1] -0.7219391
licorice_30_min_pain_g_computation$result$estimate[1] # G-computation
## [1] -0.7219391
coef(licorice_30_min_pain_glm_adjusted)["arm1. 0.5g Licorice"] # ANCOVA
## arm1. 0.5g Licorice
## -0.7219391
```

■ ANCOVA/G-Computation Match; Similar to t-test: Standard Errors smaller

```
licorice_30_min_pain_t_test$stderr # Unadjusted SE
## [1] 0.1566277
ate_i <- # ANOVA: Robust SE
  which(names(coef(licorice_30_min_pain_glm_adjusted)) == "arm1. 0.5g Licorice")
sqrt(licorice_30_min_pain_ancova_robust_vcov[ate_i, ate_i])
## [1] 0.1510823
licorice_30_min_pain_g_computation$result$se[1] # G-computation: Robust SE
## [1] 0.1510823
```

Computing Relative Efficiency

- How much more precise is adjusted analysis? $SE(\hat{\theta}) = \sqrt{var(\hat{\theta})}$
- Precision is the reciprocal of variance

```
var_unadj <- (licorice_30_min_pain_t_test$stderr)^2
var_adj <- (licorice_30_min_pain_g_computation$result$se[1])^2

prec_unadj <- 1/var_unadj
prec_adj <- 1/var_adj

prec_adj/prec_unadj # Improvement in Precision
## [1] 1.074756
1 - (var_adj/var_unadj) # Reduction in variance
## [1] 0.06955598
```

- This works for any type of analysis: getting CI more difficult

Missing Data

Taxonomy of Missing Data: (Rubin 1976)

- $R_i := 1$ if Outcome Y_i observed; 0 otherwise - Completeness
- Missingness Mechanism: How does probability of being observed relate to the data at hand?
- Missing completely at random (MCAR): $Pr\{R|Y, A, X\} = Pr\{R\}$
 - Unrelated to baseline status, treatment assignment: **Testable**
- Missing at random (MAR): $Pr\{R|Y, A, X\} = Pr\{R|A, X\}$
 - Only related to what we observe: Least restrictive assumption that does not involve unobserved data
- Missing Not At Random (MNAR): Missingness depends on data that is not observed
 - Dropout: MAR/MCAR: Data can't distinguish - Sensitivity analysis

Covariate Adjustment & Missing Data

- Guidance on missing data: ([Benkeser et al. 2020](#); [Morris et al. 2022](#))
- Covariates: Single imputation or Multiple Imputation, but make sure imputation preserves $A \perp X$
- Complete case unadjusted analyses only valid if outcomes are MCAR.
- G-computation: outcomes are MCAR **or** model correctly specified
- Doubly robust methods: Model attrition process - Correct bias induced by outcomes which are missing at random.
 - Consistent estimate if outcomes are MAR and either outcome model or missingness model is correctly specified ([Bang and Robins 2005](#)).
 - Augmented Inverse Probability Weighted (AIPW) Estimator, Targeted Maximum Likelihood Estimator (TMLE), Doubly Robust Weighted Least Squares (DRWLS)

Doubly Robust Weighted Least Squares (Robins et al. 2007)

- 1 **Fit a missing data model** predicting complete (1) vs. missing (0) outcome: off-study observations: $E[R|A, X] = g_R^{-1}(X, A; \beta_R)$
 - Logistic Regression: Completeness indicator R on A, X
- 2 **Create a weight for each individual** that is the inverse of their predicted probability of being observed: $w_i = 1/\hat{R}_i = 1/g_R^{-1}(\hat{\beta}_{0(R)} + \hat{\beta}_{0(A)}A + \hat{\beta}_{X(R)}X)$
 - Give greater weight to complete cases
- 3 **Fit a weighted GLM for the outcome** using a canonical link among those with complete data: $E[Y|A, X, R = 1]$
- 4 **Predict** the outcome for each randomized individual i under each treatment assignment $a \in \{0, 1\}$, denoted $\hat{Y}_i^{(a)}$
- 5 **Average** these predictions over the sample: $\hat{\mu}^{(a)} = (1/n) \sum_{i=1}^n \hat{Y}_i^{(a)}$
- 6 **Contrast** the average predictions: Difference, Ratio

drwls::drwls - Code

```
licorice_30_min_pain_drwls <-  
  drwls::drwls(  
    outcome_formula = # Treatment-Stratified: List of Formulas  
    list(  
      pacu_30_min_throat_pain ~ age_bl + gender +  
        asa_physical_status_bl + bmi_bl + mallampati_bl,  
      pacu_30_min_throat_pain ~ age_bl + gender +  
        asa_physical_status_bl + bmi_bl + mallampati_bl  
    ),  
    outcome_family = gaussian,  
    treatment_column = "tx",  
    missing_formula = ~ tx + age_bl + bmi_bl + mallampati_bl,  
    estimand = "difference", # Compute Difference in Means  
    data = licorice_gargle,  
    se_method = "asymptotic", # "bootstrap"  
    # Use Tsiatis (2008) Small Sample Variance Correction  
    variance_adjustment = variance_adjustment_tsiatis  
  )
```

drwls::drwls - Results

```
##
## Estimand: Difference in Means
## Method: Doubly-Robust Weighted Least Squares
## Outcome Models: treatment-stratified gaussian GLMs with an identity link
## Censoring Model: a binomial GLM using a logit link
##
##           Estimand   Estimate      SE      LCL      UCL      Z      p-value
## 1 E[Y|A=1] - E[Y|A=0] -0.7141243 0.1547758 -1.017479 -0.4107694 -4.613929 3.951273e-06
## 2           E[Y|A=1]  0.2803998      NA      NA      NA      NA      NA
## 3           E[Y|A=0]  0.9945241      NA      NA      NA      NA      NA
##
## Variance Estimation: Sandwich Estimator with Tstiatitis et al. (2008) adjustment
## CI Method: Wald 95% CI with Tstiatitis et al. (2008) adjustment
## Details:
## Outcomes: (N = 233/235)
## Outcome Formula - A=1 (N = 117/118): pacu_30_min_throat_pain ~ age_bl + gender + asa_physical_status_bl +
## Outcome Formula - A=0 (N = 116/117): pacu_30_min_throat_pain ~ age_bl + gender + asa_physical_status_bl +
## Missing Formula: !is.na(pacu_30_min_throat_pain) ~ tx + age_bl + bmi_bl + mallampati_bl
```


drwls::drwls - Risk Ratio

```
licorice_30_min_any_pain_rr_drwls <-  
  drwls::drwls(  
    outcome_formula = # Treatment-Stratified: List of Formulas  
    list(  
      pacu_30_min_any_throat_pain ~ age_bl + gender +  
        asa_physical_status_bl + bmi_bl + mallampati_bl,  
      pacu_30_min_any_throat_pain ~ age_bl + gender +  
        asa_physical_status_bl + bmi_bl + mallampati_bl  
    ),  
    outcome_family = gaussian,  
    treatment_column = "tx",  
    missing_formula = ~ tx + age_bl + bmi_bl + mallampati_bl,  
    estimand = "ratio", # Compute Relative Risk  
    data = licorice_gargle,  
    se_method = "bootstrap", # Delta method not yet implemented  
    # Use Tsiatis (2008) Small Sample Variance Correction  
    variance_adjustment = variance_adjustment_tsiatis  
  )
```

drwls::drwls - Risk Ratio Results

```
##
## Estimand: Ratio of Means
## Method: Doubly-Robust Weighted Least Squares
## Outcome Models: treatment-stratified gaussian GLMs with an identity link
## Censoring Model: a binomial GLM using a logit link
##
##           Estimand  Estimate      SE      LCL      UCL      Z p-value
## 1 E[Y|A=1]/E[Y|A=0] 0.5525840 0.13216634 0.3114471 0.9080484 -3.385249      NA
## 2           E[Y|A=1] 0.1946247 0.03776607 0.1222978 0.2939994      NA      NA
## 3           E[Y|A=0] 0.3522083 0.04605799 0.2535551 0.4603782      NA      NA
##
## Variance Estimation: Non-parametric bootstrap using 10000 replicates with Tstiatis et al. (2008) adjustment
## CI Method: Bias Corrected and Accelerated (BCa) 95% CI with Tstiatis et al. (2008) adjustment
## Details:
## Outcomes: (N = 233/235)
## Outcome Formula - A=1 (N = 117/118): pacu_30_min_any_throat_pain ~ age_bl + gender + asa_physical_status
## Outcome Formula - A=0 (N = 116/117): pacu_30_min_any_throat_pain ~ age_bl + gender + asa_physical_status
## Missing Formula: !is.na(pacu_30_min_any_throat_pain) ~ tx + age_bl + bmi_bl +      mallampati_bl
```

Why use drwls Package?

- Continuous, Binary, Count outcomes: `outcome_family`
- Doubly Robust: `drwls::drwls()` - Formula for missingness model
 - MAR + either outcome or missingness model correctly specified
- Stratified Models/Treatment-covariate interactions: Improve precision (Ye et al. 2022)
- Impute missing covariates: Mean/Mode imputation; Can use `mice::mice`
- Bootstrap or Sandwich variance estimation
- Small sample size degree of freedom corrections (Tsiatis et al. 2008)

In Sum: Hands-On

- HTML file - Examples: Binary, Ordinal, Time-to-Event
- **R** has broadest range of methods; Room for improvement in usability; Software planning & validation
 - Syntax largely follows regression: Additional arguments
 - Off-the-shelf software; Macros; Dedicated packages
- On-line tutorials and data resources: Learning & Teaching
- Can use estimates of relative efficiency with historical data

FAQ

Learning Objectives: FAQ

- Identify common points of confusion about covariate adjustment and how to clarify these issues to the study team.
 - Role of Observed Balance in Precision Gain
 - Selecting covariates
 - Efficiency Guarantees
 - Missing covariates
 - Longitudinal or Clustered Data; Cluster Randomized Trials

FAQ

Questions About Covariate Adjustment

- Where to start? (Van Lancker, Bretz, and Dukes 2024; Bannick, Yi, and Ye 2026; U.S. Food and Drug Administration 2023; Benkeser et al. 2020)
- How to decide which variables to include?
 - Prognostic; Pre-specified *a priori*; Data from similar population
 - Data adaptive: Active research; Large trials; Crossfitting
- Should adjustment be guided by observed balance?
 - Change in Point estimate: Association & Balance
 - Efficiency: Association & *Variance in imbalance*: Still gain if balanced
- Can covariates be correlated? Is multicollinearity an issue?
 - Prediction not interpretation; Not testing coefficients
 - Correlated: Each covariate has incremental prognostic value

Questions About Covariate Adjustment

- Can covariate adjustment reduce precision?
 - Guarantees asymptotic; Not all methods have guarantees ([Colantuoni and Rosenblum 2015](#)); Weak/null association: increase variance
 - Corrections: CI Coverage; # of parameters ([Tsiatis et al. 2008](#))
- Can covariate adjustment be beneficial in large trials?
 - Yes: *Absolute* imbalance decrease with N , *standardized differences* do not ([Senn 1989](#)). Imbalance is an issue for any size trial.
 - Larger trial: same precision gain = larger absolute reduction in N
- Can adjustment leverage post-randomization covariates?
 - Selection Bias; Missingness/Censoring: Plausibility; Specific Methods

Questions About Covariate Adjustment

- Can covariate adjustment play nicely with others?
 - ✓ Covariate Adaptive Randomization: Variance Estimation
 - ✓ Group Sequential Design: Orthogonalization ([Van Lancker, Betz, and Rosenblum 2025](#))
- How should covariate adjustment be used in design calculations?
 - Gain: Magnitude, Precision, Transportability ([Siegfried, Senn, and Hothorn 2022](#))
 - Rarely precisely known *a priori*: Transportability always a question
 - Conservative: Unadjusted; *Insurance* against other factors
 - Information Monitoring: Adaptive trial design - Reduce sample size ([Mehta and Tsiatis 2001](#); [Van Lancker, Betz, and Rosenblum 2025](#))

Questions About Covariate Adjustment

- Adjust for individual variables or prognostic scores? PROCOVA
 - PROCOVA = ANCOVA: prognostic modeling to create “super-covariate”
 - Assumes no treatment-covariate interactions; May be less efficient
 - Still faces issues of imprecision and transportability
- How to address missing data in covariates?
 - Methods uphold independence of treatment and covariates
 - Exclude covariates with lots of missing data, Mean/Mode imputation ([Benkeser et al. 2020](#))
 - More complex imputation may improve efficiency: $A \perp\!\!\!\perp X_{\bar{S}}$; Subtleties: When to apply Rubin’s Rules? ([Morris et al. 2022](#))
 - Doubly Robust methods ([Bang and Robins 2005](#); [Kang and Schafer 2007](#))

Questions About Covariate Adjustment

- Can covariate adjustment be used in longitudinal analyses? Yes - ANHECOVA can be generalized to MMRM framework ([Wang and Du 2023](#))
- Can covariate adjustment be used in cluster randomized trials?
 - Yes, even designs with stratified randomization, minimization, and restricted randomization ([Hemming et al. 2025](#); [Li et al. 2015](#); [Wang et al. 2026](#))
 - Addresses unique challenge: Post-randomization Selection Bias
 - Cluster RCTs: other issues & considerations ([Kahan et al. 2022](#); [Hemming and Taljaard 2023](#))

Questions About Covariate Adjustment

- Can you use machine learning for augmentation terms?
 - Yes: Bias/Variance Tradeoff; Data Adaptive Approaches
 - Large Trials & Non-linear Associations
- General Guidance on Approach: ([Shao et al. 2026](#))
 - In general, simple models and selection strategies worked best
- How much precision can you gain?
 - Depends: Correlation between baseline and outcomes; Transportability of association to new trial ([Siegfried, Senn, and Hothorn 2022](#))
 - Higher end: 20-25% with good prognostic scores

In Sum: FAQ in Practice

- 1 Choose covariates *a priori* - natural history: Correlated but contain *independent prognostic information* (prediction not interpretation): partial R^2 . Low missingness - Imputation preserves $A \perp X$. Weak/noisy predictors can reduce precision.
- 2 Efficiency gains depends on the prognostic value and *the variance of the imbalance*, not the observed imbalance.
- 3 The difference between unadjusted and adjusted estimates depends on the prognostic value and the observed imbalance. Analyses shouldn't be guided by observed imbalance.
- 4 Precision gains still important in large studies: same gain means larger proportional reduction in N .
- 5 Bias-Variance tradeoff vs. Sample Size: Benefit of flexible models may not be realized at feasible sample sizes.
- 6 Best to design trials assuming no gain (imprecision, transportability): Use flexible study designs to reduce sample size.

References

Learning & Using R

- R For Data Science, 2nd Ed
- R Studio Education
- R Studio Cheatsheets
- R Studio Books
- Awesome R Learning Resources
- pharmaverse
- Hands-On Programming with R
- R Programming for Data Science
- R Markdown: The Definitive Guide
- Advanced R
- Quarto
- swirl
- Modern Statistics with R
- The R Warehouse
- Happy Git with R
- Posit Recipes
- Data Carpentry
- Reddit: [r/Rlanguage](#) and [r/Rprogramming](#)
- R Packages 2nd Ed
- Coursera Data Science Foundations in R
- CRAN Task Views

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